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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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TREND ANALYSIS OF SOCIAL MEDIA ENGAGEMENT USING EXPLORATORY DATA ANALYSIS

DSA0404 – Fundamentals of Data Science

Date: 04.09.2026

Batch: 5

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Introduction

- Social media platforms generate large amounts of engagement data every day.
- Engagement includes likes, comments, shares, and views.
- Exploratory Data Analysis (EDA) helps identify patterns and trends in this data.
- This project analyzes engagement trends and presents them through visual dashboards.





Objectives



- Collect and preprocess social media engagement data.
- Perform Exploratory Data Analysis (EDA) to identify trends and patterns.
- Compare engagement across different social media platforms and content types.
- Visualize key engagement metrics using interactive charts and graphs.
- Identify high-performing posts and optimal posting trends.
- Develop an interactive dashboard for easy analysis and reporting.

Literature Review



S.No	Title	Technique	Year	Problem Addressed
1	Social Media Analytics Using EDA	EDA	2024	Difficulty in identifying engagement trends
2	Instagram Engagement Analysis	Data Visualization	2024	Manual analysis of engagement
3	Social Media Performance Dashboard	Dashboard Analytics	2025	Lack of visual insights
4	Engagement Prediction in Social Media	Machine Learning	2025	Low prediction accuracy



Existing System



- Social media engagement data is analyzed manually or through basic reports.
- Limited visualization makes it difficult to identify trends.
- Comparing engagement across different platforms is time-consuming.
- Existing methods provide limited insights for decision-making.
- Lack of an interactive dashboard for effective analysis.



Proposed System



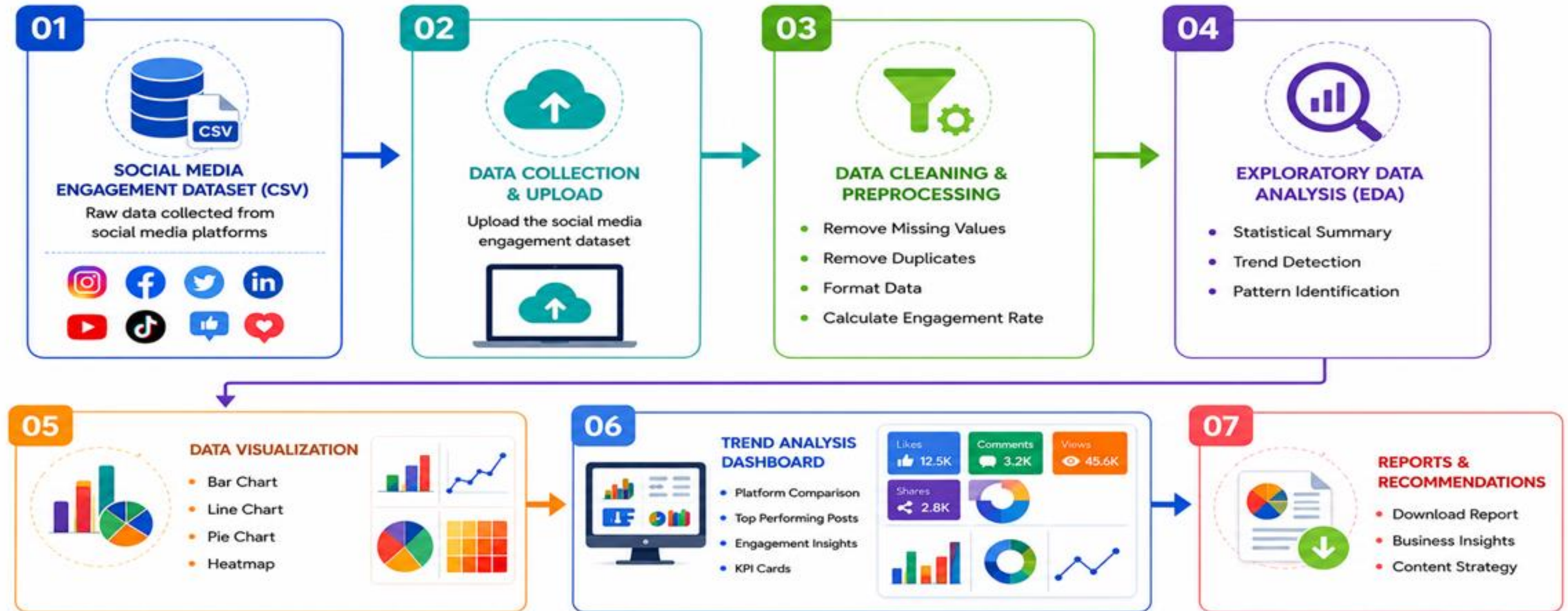
- Collect and preprocess social media engagement data automatically.
- Perform Exploratory Data Analysis (EDA) to identify trends and patterns.
- Visualize engagement using interactive charts and graphs.
- Compare engagement across different platforms and content types.
- Display insights through an interactive dashboard for better decision-making.



Engineer to Excel



System Architecture





Module 1 – Data Collection and Preprocessing



Purpose:

To collect and prepare raw social media engagement data for analysis.

Working:

- Collect data from public datasets and APIs.
- Remove duplicates and handle missing values.
- Standardize timestamps and transform data.
- Generate features like engagement rate, posting hour, day, and content length.

Output:

A clean and structured dataset ready for analysis.



Module 2 – Exploratory Data Analysis and Trend Visualization



Purpose:

To analyze the cleaned data and identify important engagement patterns and trends.

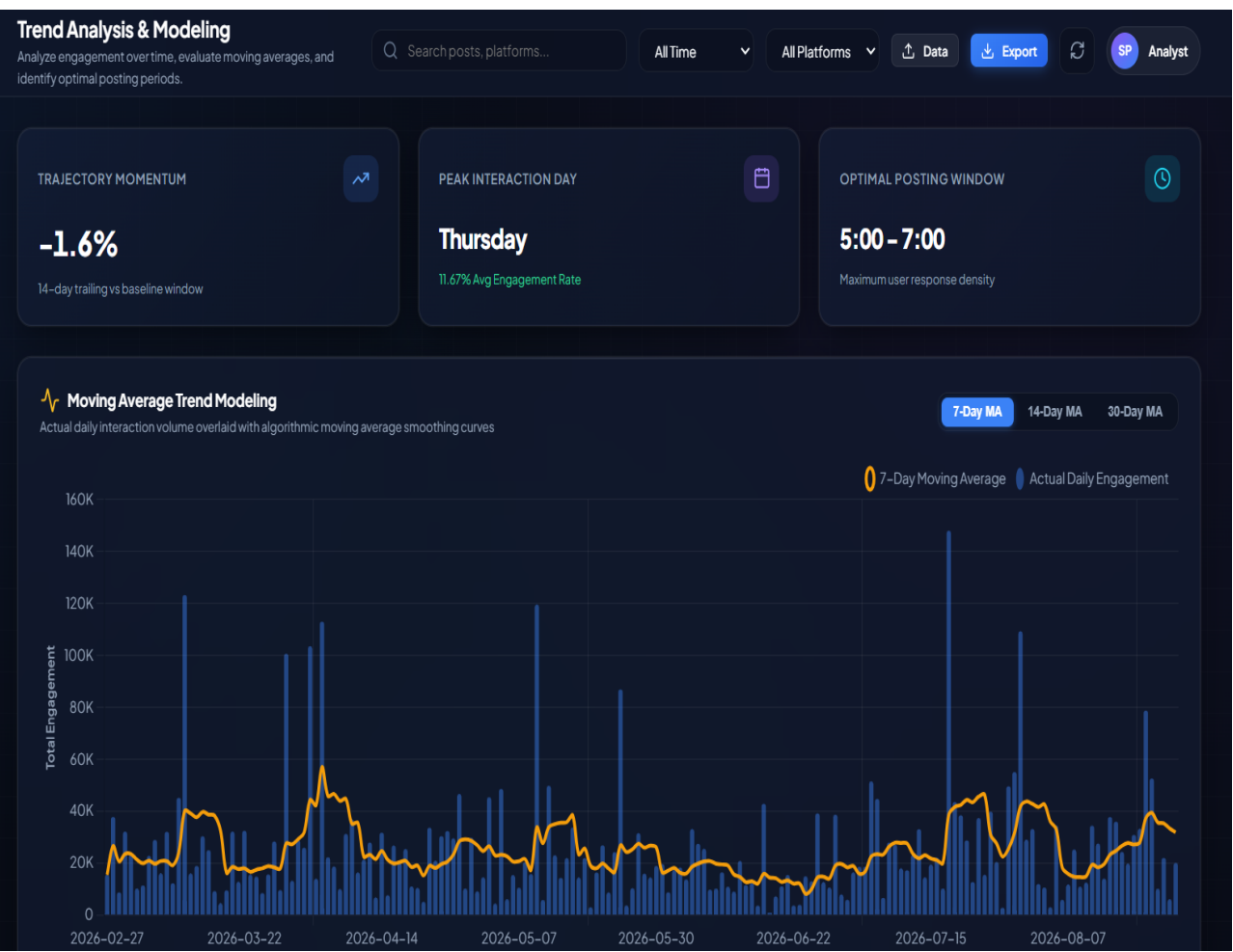
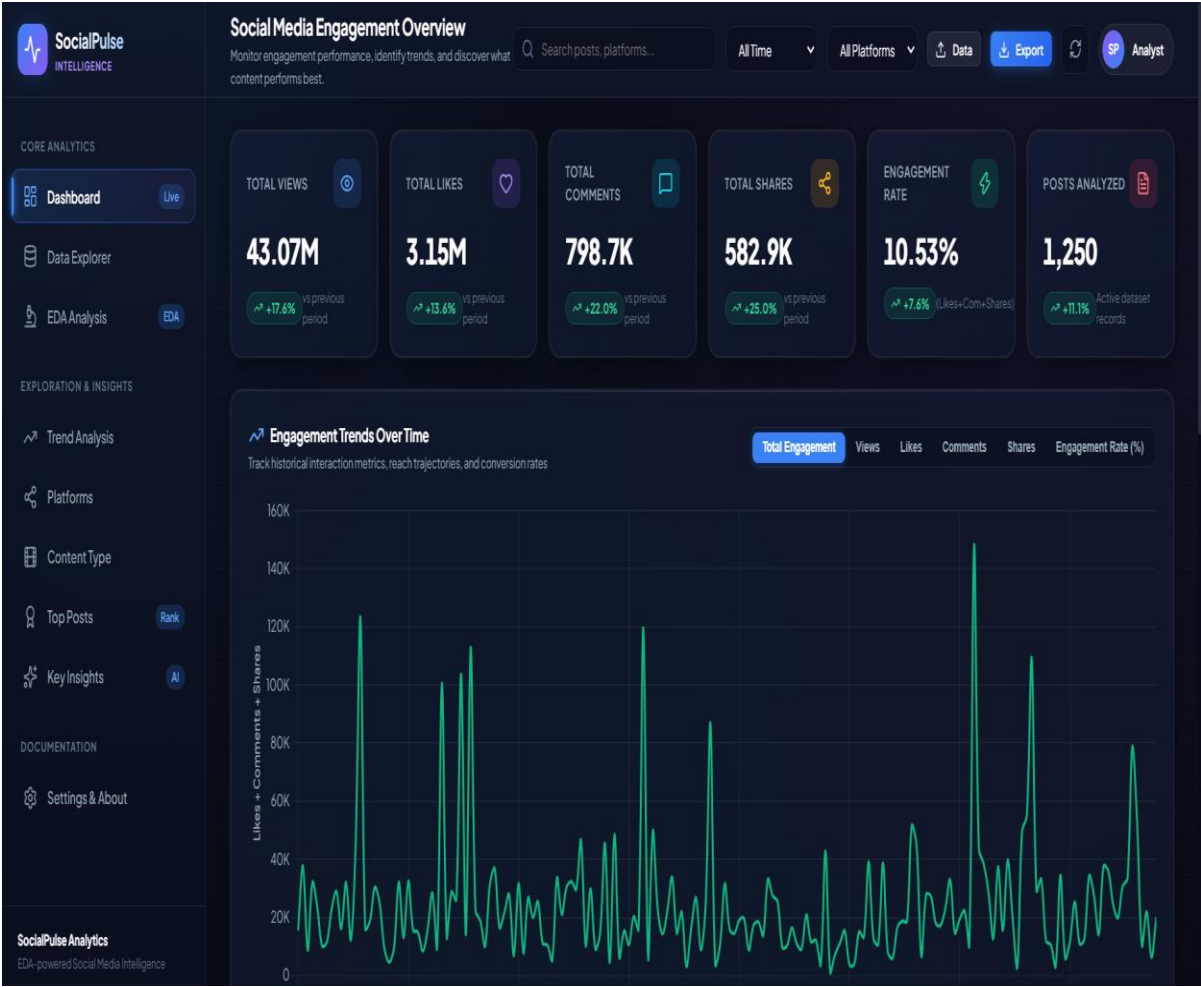
Working:

- Perform descriptive and distribution analysis.
- Analyze relationships using correlation analysis.
- Compare engagement across platforms and content types.
- Visualize trends using charts and graphs.

Output:

Clear statistical and visual insights into social media engagement trends.

Implementation



Implementation

Top Performing Posts

Identify viral breakout posts and benchmark individual post metrics against baseline averages.

Search posts, platforms...

All Time

All Platforms

Data

Export

Refresh

SP

Analyst

Platform Comparison

Benchmark reach, likes, comments, and engagement rates across social platforms.

Search posts, platforms...

All Time

All Platforms

Data

Export

Refresh

SP

Analyst

Post Discovery & Filtering

Refine top posts by channel, format, and threshold cutoffs

Reset Filters

Platform

Content Format

Min Views Threshold

Rank Sorted By

All Platforms

All Formats

Any Views

Total Engagement (Default)

Ranked Top Posts

Displaying top 30 of 1,250 matching posts

RANK	POST ID	TITLE / CAPTION	PLATFORM	TYPE	DATE	VIEWS	LIKES	COMMENTS	SHARES	ENGAGEMENT RATE
1	POST-00800	Mastering user retention throug...	TikTok	Reel/Short	2026-07-18	538,344	68,777	17,471	28,186	21.26%
2	POST-00624	Deep dive into multivariate stati...	YouTube	Video	2026-03-12	864,933	77,326	28,247	3,418	12.6%
3	POST-00158	Why your engagement rate is dr...	TikTok	Reel/Short	2026-04-02	344,876	70,379	11,618	14,682	28.03%
4	POST-00355	Top 5 Python libraries for Explor...	YouTube	Live	2026-04-04	468,346	61,883	20,235	4,534	18.5%
5	POST-00739	Behind the scenes: redesigning ...	TikTok	Reel/Short	2026-05-10	303,002	53,422	12,435	12,070	25.72%
6	POST-00982	Unlocking higher conversion rat...	Instagram	Reel/Short	2026-07-30	320,918	56,544	10,592	9,660	23.93%
7	POST-00530	Why consistency beats viral spik...	TikTok	Reel/Short	2026-05-24	432,843	49,183	8,514	18,239	17.54%

TikTok

298 posts

13.97%

Avg ER • 13.45M views

YouTube

289 posts

8.63%

Avg ER • 17.96M views

Instagram

338 posts

11.06%

Avg ER • 6.99M views

Facebook

138 posts

6.36%

Avg ER • 1.83M views

X/Twitter

142 posts

7.46%

Avg ER • 2.48M views

LinkedIn

45 posts

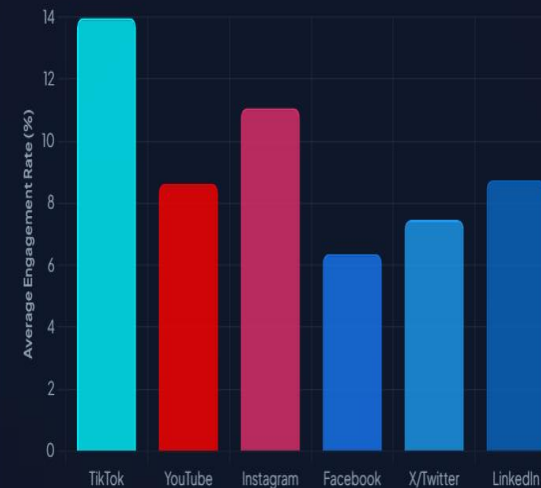
8.74%

Avg ER • 354.5K views

Platform Performance Metric Comparison

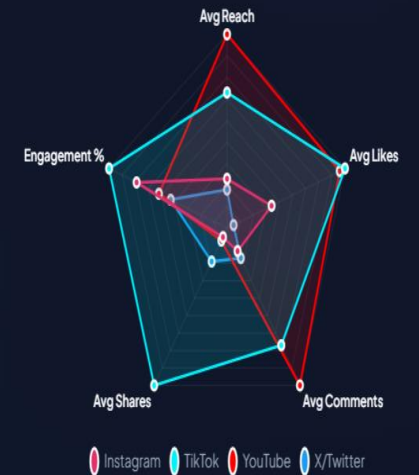
Cross-channel reach and engagement rate ranking

Engagement Rate (%)



Multi-Dimensional Platform Radar

Normalized score comparison across 5 core engagement vectors





Conclusion

- The project analyzes social media engagement data using Exploratory Data Analysis (EDA).
- It identifies engagement trends, popular content, and user interaction patterns.
- Interactive visualizations make the analysis simple and easy to understand.
- The dashboard provides meaningful insights for better content planning and decision-making.
- This project demonstrates how data science techniques can transform raw social media data into valuable business insights.

REFERENCES

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THANK YOU